

Protocol-aware wind-power event measurement

China

4 archives · 298 turbines

Europe

4 archives · 42 turbines

Physical time · quality flags · training-only scaling

1 Detection



Ramps · V / inverted V
60–240 min

2 Correspondence



One-to-one + overlap
IoU · coverage

3 Representation



Raw / PCA / angular
polarity + scale

4 Evaluation and decision

Structural survival

ARI · NMI · coverage
Split / merge · composite

Physical tracking

LiDAR · WindCube · ERA5
Endpoint + full trajectory

Forecast-to-cost

Issued forecast → rule charge
Debit · credit · storage

Independent references

Blind ratings · weather · LiDAR · prices